SA 220 AQUACULTURE

HYPOPHYSATION-CATFISH

CATFISH HYPOPHYSATION

- **Introduction:**

- Availability of quality and quantity seed remains of the key challenges hindering sustainable aquaculture.

- Lack of expertise and necessary training in hatchery management and techniques are some of the issues affecting the availability of quality and quantity of seed.

- The situation is even compounded by some species such Catfish having challenges to reproduce in captivity.

CATFISH HYPOPHYSATION

- **Introduction:**

- To provide consistent fast growing, disease resistant and uniform sized catfish fingerlings and juveniles, there has been investments in catfish hatcheries run by individuals or governments.

- Artificial propagation by induced breeding through hormone treatment, followed by artificial fertilization and incubation of fertilized eggs and subsequent rearing up to fingerling size has been found to have several advantages.

CATFISH HYPOPHYSATION

- **Introduction:**

What is Artificial Propagation?

- This is a chain of activities which are more or less similar to those of natural reproduction.

- This involves the intervention of man in the male and female fish reproductive products (called the milt and eggs respectively) coming together to form fertilized eggs and subsequently forming fingerlings, juveniles and ultimately adult fish.

CATFISH HYPOPHYSATION

- **Key advantages of artificial propagation are;**

1. Increase fertilization and hatching rates
2. Provision of protection against enemies
3. Provision of favourable conditions for growth and survival

- **Some definitions (revision)**

1. Hatchery: A place for artificial breeding. In the case of fish, eggs are hatched under artificial conditions and rearing through the early life stages (eggs – larva – fries – fingerlings/juveniles)

CATFISH HYPOPHYSATION

2. Hypophysation is artificial or induced breeding, a technique that uses pituitary gland extract injection for breeding fish.

3.Brood-stock: Brood-stock or Brood-fish are a group of mature individuals used in aquaculture for breeding (reproduction).

FACTORS FOR SUCCESSFUL HATCHING

- The success of artificial propagation depends on the **size of the female gonad** and **gonadal maturity**.

CATFISH HYPOPHYSATION

FACTORS FOR SUCCESSFUL HATCHING

- Availability of good water quality is very essential
- Good parent stock (brood fish) quality from a reliable source
- Proper calculation and administration of hormone dosage.
- Proper testes collection and sperm preparation.
- Accurate determination of the time of ovulation in females

CATFISH HYPOPHYSATION

FACTORS FOR SUCCESSFUL HATCHING

- Good stripping and fertilization techniques.
- Quality feed and timely feeding period.
- Intensive record keeping.
- Bio-security

How to Identify an adult male and female catfish

1. Size

Males are smaller and narrower than females.

2. Size of the Head

The head of the male is larger and wider than the female.

3. Width of the body

Females are wider than males in the belly area; the extra room or space is needed to carry many eggs.

How to Identify an adult male and female catfish

4. Dorsal Fins

Males have longer fins than that of Females as well as males having brighter colored fins.

5. Openings (Genital)

Both have two openings at the lower belly, one for the anus and the other for the genitals.

Right before spawning (eggs are produced), the female genital area will be reddish and swollen

How to Identify an adult male and female catfish



Fig. 3: **Female:** the genital area is flat and has an opening that is rounded

Male: the genital are raised and shaped like a nipple

Criteria for selecting brood-stock-females

1. Choose only female fishes that weigh above 1kg

   and must be a year and above old.

WHY?





Criteria for selecting brood-stock-females

Because they have a substantial quantity of mature eggs and are easy to manipulate.





Criteria for selecting brood-stock-females

2. Female fish should have a well-rounded and soft abdomen.



3. Eggs, showing clearly the nucleus in the centre.

4. Genital opening is swollen and sometimes reddish or rose in colour.





Criteria for selecting brood-stock-females

5.When pressing out the female eggs for “inspection” choose only the one that have eggs that separate from each other and are golden in colour.

6. Make sure that the outlet where the egg of the female fish will come out is slightly dark in colour.

7. Make sure that the tummy of the female fish slightly protrudes out, when placed flat-faced down.

Criteria for selecting brood-stock-males

1. Choose only male fishes that weigh above 1kg and must be a minimum of a year old.

2. No clear external symptoms to indicate the maturity

3. Elongated and slender in body shape

4. Swollen urogenital papilla and the male organ “crosses” and its pink at the dot end.



Fig. 6: Features of the male breeder



Use of Hormone for Inducement

There are two methods of inducing a matured brood-stock for egg production:

1.Use of pituitary glands and

2. Use of hormone to induce the brood-stock.

The steps for injecting female broodstock

1. When more than 10 females are to be injected, separate them into two groups of equal numbers and inject them 30–60 minutes apart.

- ❑ This period is known as **latency period.**

1. When selecting gravid broodfish, weigh the female and place her on a firm surface or the work table

2. Extract the pituitary gland from either male or female fish (regardless of the species) and then store it in ethanol before using it to induce egg maturity in the selected female broodstock

The steps for injecting female broodstock

4. Choose the type of hormone

4.1 Natural hormones, such as African catfish pituitary gland, For freshly removed pituitary glands, the recommended ratio for recipient to donor is 1 kg to 1 kg.

- If dried, the recipient to donor ratio for preserved pituitary glands is 1 kg to 1.5 kg.

The steps for injecting female broodstock



3A. Cut near pelvic fin



3B.Cut near pelvic fin



3C. Make incisions behind the skull



3D. Remove gills for convenience

The steps for injecting female broodstock



3E. Sella turcica



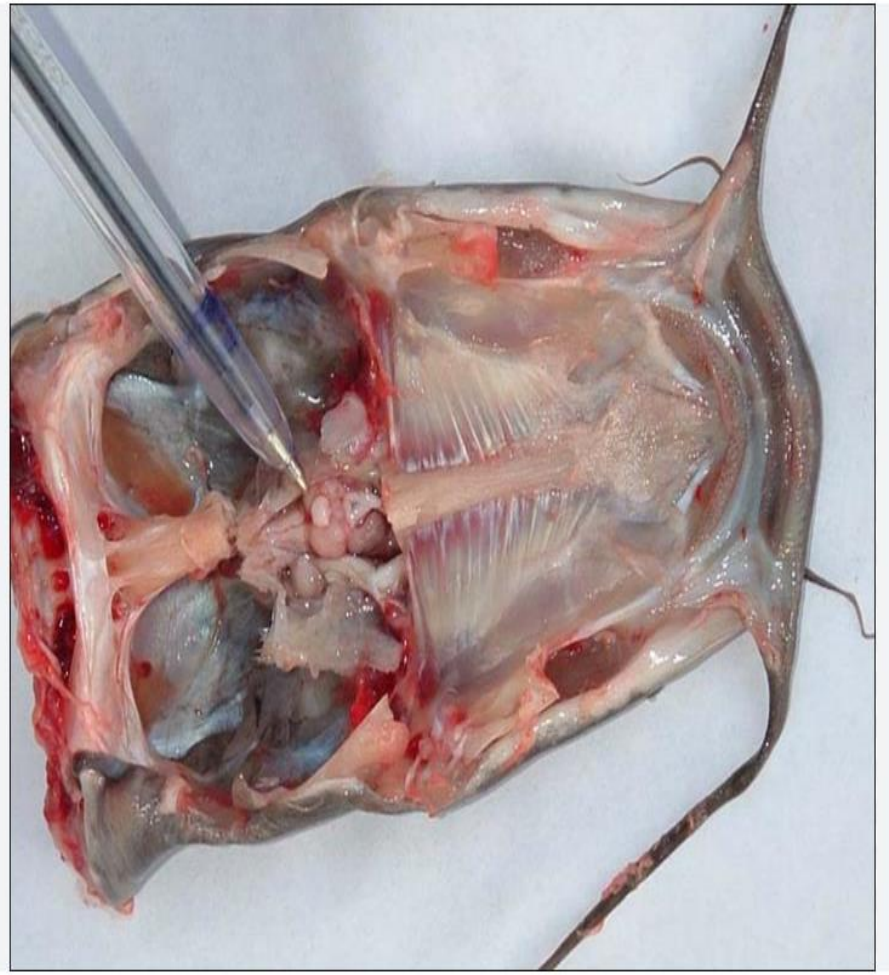
3F. Fish brain



3G. Location of Pituitary gland



3H. Pituitary glands in preservatives



Preserving the PG

There are three methods for preservation of fish pituitary glands according.

1. Preservation in absolute alcohol.

- Immediately after collection, the glands are washed and transferred to fresh absolute alcohol in amber coloured glass vials.
- Every after 24 hrs, pituitary glands are washed with absolute alcohol and stored at room temperature in a dark place to avoid the sunlight

Preserving the PG

- To increase the shelf life of the glands it is advised to refrigerate.
- During washing and storage, absolute alcohol dehydrates the glands, dissolves fats and preserves it for a longer time (up to 2 years).
- Vials are labelled for information includes date of collection, fish name, maturity stage and other details if any

Preserving the PG

2. Preservation in acetone

- Upon collection, glands are transferred to ice-chilled acetone and stored in a refrigerator for 2-3 days.
- During refrigeration, acetone is changed 2-3 times and stored in a refrigerator.
- Like absolute alcohol, acetone also has dehydrating and defattening effect and vials are labelled as explained above

Preserving the PG

3. Immediate freezing

Immediately upon collection the glands are kept in freezer.

The steps for injecting female broodstock

4. 2 Synthetic hormones, like Ovaprim or Overtide.

- The manufacturer usually gives the prescription and dosage in accordance to the weight of the female fish. For synthetic hormones, inject 0.5 ml of the hormone for a 1 kg female fish







The steps for injecting female broodstock

- Ova prim is usually in a 10 ml bottle.
- One bottle of the hormone can inject 20kg of females or about 10 females at 2kg weight or 20 femalesat 1kg each.

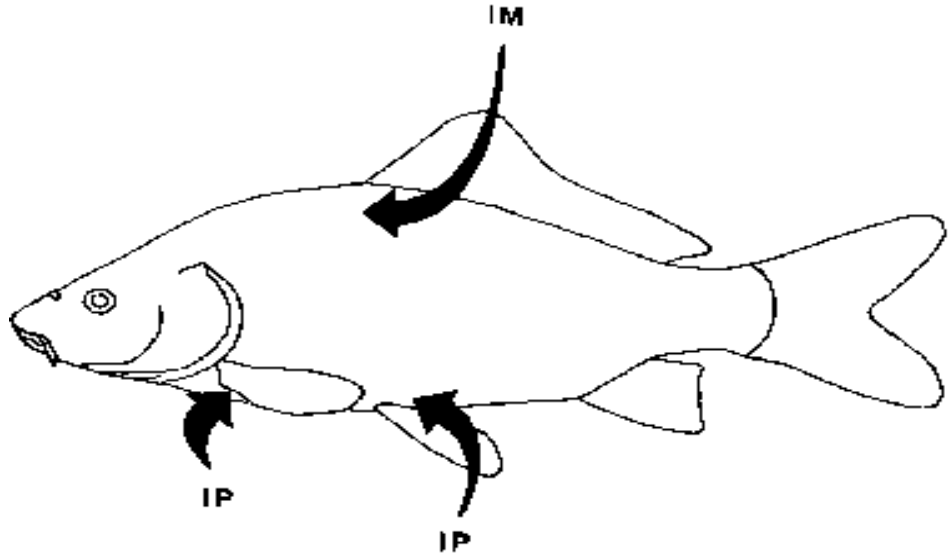






HOW TO INJECT

- Inject it just below the anterior part of the dorsal
- fin and above the lateral line by pointing the syringe toward the tail.
- Inject the fish into the muscle above the lateral line with the needle at a 30- to 45-degree angle to the body of the fish
- intramuscularly or intraperitonially



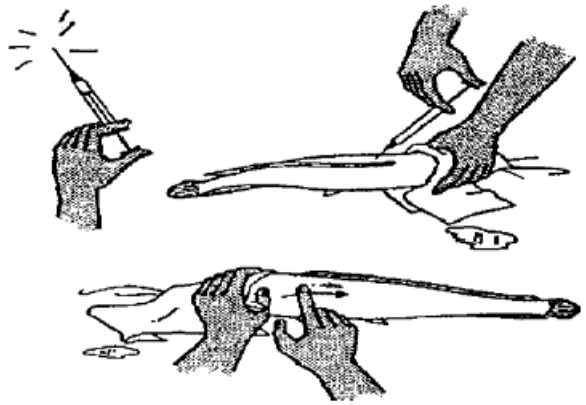


Fig.8: Syringe and how to inject the brooder-stock

HOW TO INJECT



5A. Near the ventral fin base



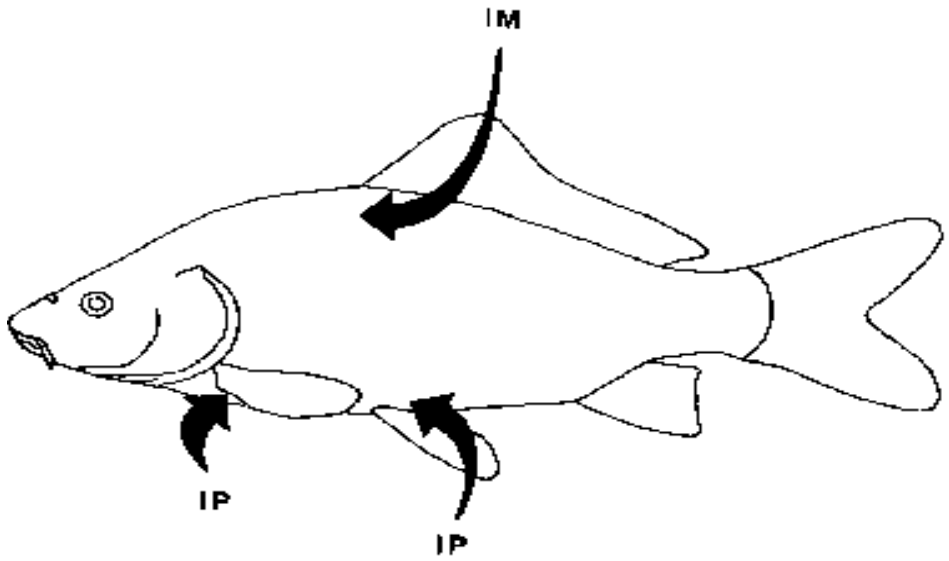
5B. Below the dorsal fin base and above the lateral line



5C. On the caudal peduncle and above the lateral line



5D. Below the pectoral fin



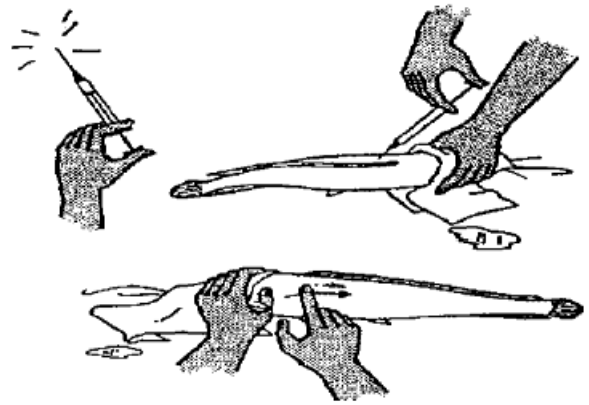
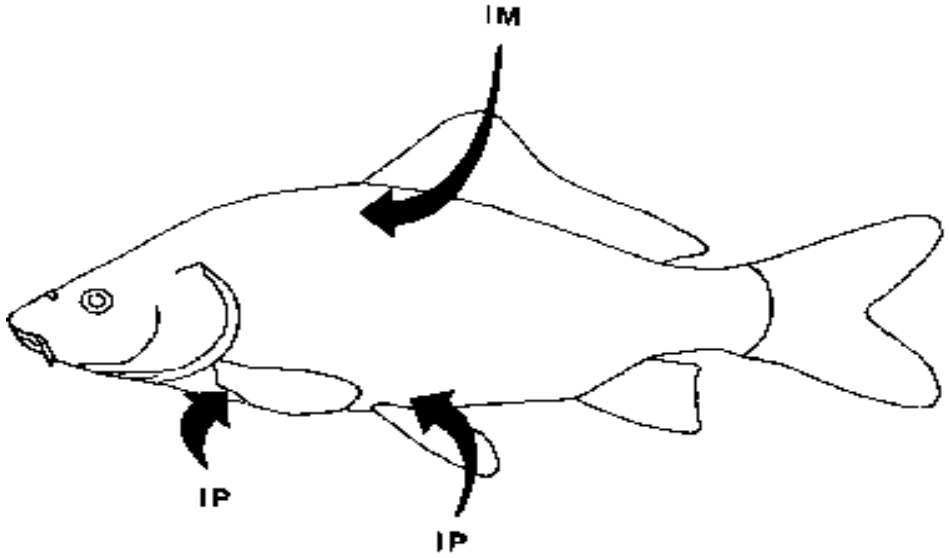


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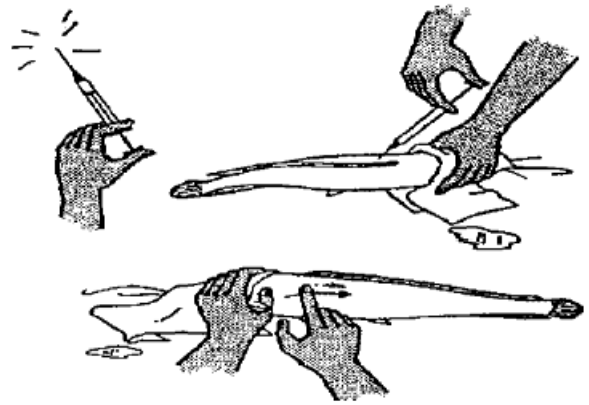


Fig.8: Syringe and how to inject the brooder-stock

HOW TO INJECT

- Carefully and gently withdraw the syringe.
- Massage the injected spot with your finger to evenly distribute the hormone into the muscles.

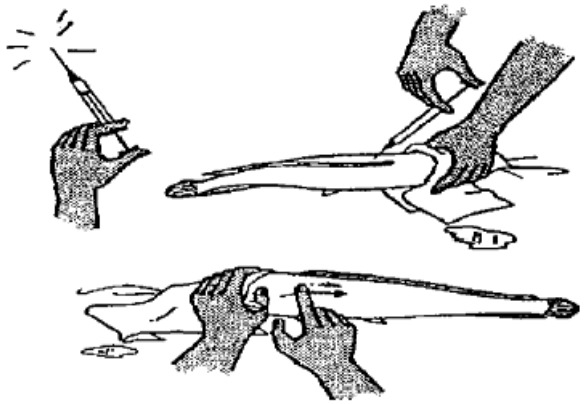
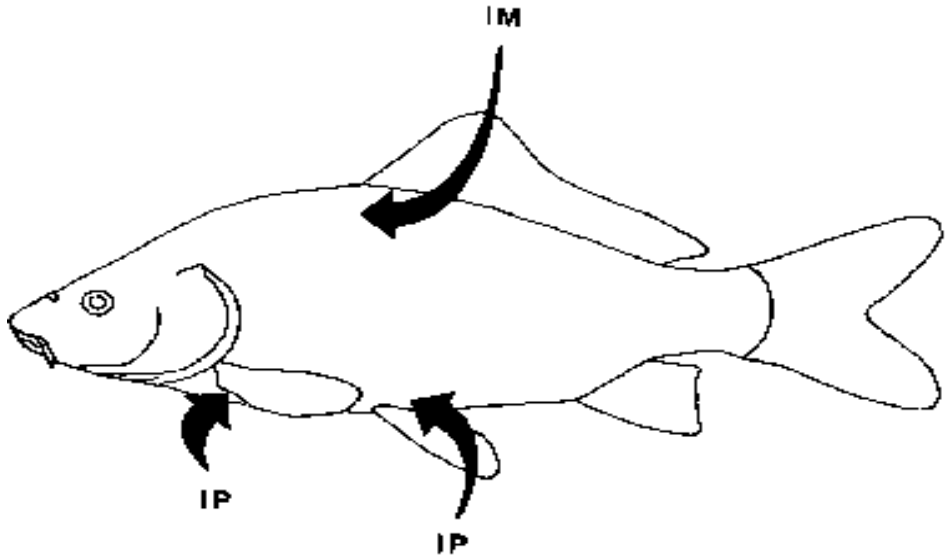


Fig.8: Syringe and how to inject the brooder-stock



The steps for injecting female broodstock

5. Put the injected female broodstock in an isolated tank or bowl of water with enough water to cover half of the body.

- Leave the fish for about 10 to 12 hours before stripping.

COLLECTION AND INCUBATION OF MILT

- About an hour before stripping, the milt must be obtained from a male brood fish.
- The milt is obtained by sacrificing one male and dissecting the testis.
- Some small incisions are made into the cream coloured lobes of the testis

COLLECTION AND INCUBATION OF MILT

- The materials to be use are: Surgical blade or a pair of scissors, container and clean cloth material.
- These materials should be sterilized and the container for milt collection rinsed properly with saline water before use.





COLLECTION AND INCUBATION OF MILT

- Use normal saline water (0.9% saline solution) for the procedure.
- A drop of a milt contains a million of active sperm cells or spermatozoa; this drop can fertilize the egg of a female if mixed properly.



INCUBATION OF MILT

- The incubation is usually done two hours before the expiration of the latency period of the female catfish.

- The procedure for the incubation of milt is as follows;



1. Cut the male open from the ventral side.

2. Detach the milt sac from the other organs of the body.



INCUBATION OF MILT

3. Use saline water to rinse sac to prevent impurity.

4. Cut the milt sac with blade or scissors into the incubating container.

5. Rinse sac with saline water.





INCUBATION OF MILT

6. Neatly wrap the container inside a bowl with cloth for a minimum of 1 hour and a maximum of 3 hours.

7. Store under room temperature for effective incubation before mixing with the egg for fertilization.

- In a situation when the milt cannot be used immediately, the milt in the saline water can be stored in a refrigerator for 24 hours before use.



STRIPPING OF BROODSTOCK

The following procedure outlines the steps required for stripping female broodstock:

1. Carefully bring the female fish out after 10–12 hours, depending on the temperature.

❑ The approximate time available for stripping, bringing about an optimal hatching percentage are;

20–30 minutes (30°C)

60–90 minutes (25°C) and

120–240 minutes (20°C)

STRIPPING OF BROODSTOCK

The following procedure outlines the steps required for stripping female broodstock:

2. Check for eggs at the base of the tank.

- If your fingers crush some eggs, this means that eggs are coming out already, which means the eggs can be said to be running. The fish is ready for stripping.

STRIPPING OF BROODSTOCK

2. Wipe the body of the broodfish with a dry soft towel.

- Two people might be required to handle the fish at both the tail and the head.

- One person holds the head with a towel while the other holds the tail





STRIPPING OF BROODSTOCK

3. Remove ripe eggs from the abdomen by gently pressing several times from the anterior end until a high percentage of the eggs is collected into a dry bowl.

- Immediately stop pressing when blood is on the eggs.

4. Carefully collect the eggs in a clean dry plastic or metallic bowl.

- Ripe eggs are usually uniform in size with a dark spot in the center, which is the nucleus.

STRIPPING OF BROODSTOCK

5. Weigh the stripped eggs to estimate the expected number fry.

- African catfish contains approximately 600 eggs per gram, so weighing the eggs helps estimate the expected number of fry for egg hatchability.

- Alternatively, the quantity of eggs can be estimated as 15%–20% of the fish's weight

EGG FERTILIZATION AND INCUBATION

The following procedure outlines the steps required for egg fertilization and incubation:



1. Spread the collected milt on the eggs in a plastic bowl

2. Immediately add the saline solution (0.9%) to the mixture.



- This will enhance the motility of the milt cells in fertilizing the eggs
- The saline water also makes the milt last longer before it fertilizes the eggs.

EGG FERTILIZATION AND INCUBATION

3. Saline water extends the lifespan of the sperm cells.

- This makes it possible for all the eggs to receive active milt.

4. Adding fresh water reactivates and makes the sperm cells motile and initiates the external fertilization process

5.Continue mixing to prevent the eggs from sticking or clogging together.

EGG FERTILIZATION AND INCUBATION

6. After fertilization, incubate the eggs in an incubating unit.

7. Spread the eggs evenly on a spawning mat with a mesh size of 1.2

x 1.2 mm inside an incubating trough (80–100 L) prefilled with water.

EGG FERTILIZATION AND INCUBATION

8. Depending on the water temperature, which should be higher than 25°C, the eggs will hatch between 20 and 36 hours and the fry will emerge.

9. Remove the hatching tray after a maximum of 36 hours to prevent fungi from attacking the fry, as unhatched eggs will attract the growth of fungi

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HOMEWORK

1.De Leeuw, R., Goos, H.J. Th., Richter, C.J.J. and Eding, E.H., 1985 Pimozide - LHRHa induced breeding of the African catfish, Clarias gariepinus (Burchell). Aquaculture, 44: 295–302.

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HOMEWORK

4. Fregene, B. T., P. Bolorunduro, R. Yossa, H. C. Karisa, A. A. Olaniyi, and I. Ajose. "Extension manual on production of quality catfish seed." (2021).

5. Amachree, Dokuboba, O. A. Akinrotimi, and P. O. Dabor. "Effects of Different Physiological Saline Concentrations on the Reproductive Performance of Clarias gariepinus (Burchell, 1822)." *Journal of FisheriesSciences. com* 12.4 (2018): 4-8.

END

THANK YOU FOR YOUR ATTENTION